

Discussion 7: The Lorentz Group – Solutions

Names:

Section:

Congratulations, you've made it halfway through the course! This week's discussion will tie together all of the mathematical and physical topics we've covered up until now. We'll clarify the relationship between rotations and Lorentz transformations by introducing a useful concept called rapidity and by considering infinitesimal Lorentz transformations. Most importantly, we will encode the constancy of the speed of light in the mathematical statement that *the Lorentz group* $SO(3,1)$ is the set of linear transformations with determinant 1 that preserve the invariant interval.

Part 1: Boosts and rapidity

For this part, we will need the definitions of the hyperbolic trig functions, $\sinh$ (pronounced “sinsh”), $\cosh$ (pronounced like it sounds), and $\tanh$ (pronounced “tansh”):

$$\sinh \phi = \frac{e^\phi - e^{-\phi}}{2}, \quad \cosh \phi = \frac{e^\phi + e^{-\phi}}{2}, \quad \tanh \phi = \frac{\sinh \phi}{\cosh \phi} = \frac{e^\phi - e^{-\phi}}{e^\phi + e^{-\phi}}. \quad (1)$$

- 1.1. The hyperbolic trig functions obey similar identities to the ordinary trig functions, but with some extra minus signs. Prove the identities

$$\cosh^2 \phi - \sinh^2 \phi = 1, \quad \cosh^2 \phi + \sinh^2 \phi = \cosh(2\phi). \quad (2)$$

First let's compute the squares of the hyperbolic trig functions:

$$\cosh^2 \phi = \left(\frac{e^\phi + e^{-\phi}}{2} \right)^2 = \frac{e^{2\phi} + e^{-2\phi} + 2}{4}, \quad \sinh^2 \phi = \left(\frac{e^\phi - e^{-\phi}}{2} \right)^2 = \frac{e^{2\phi} + e^{-2\phi} - 2}{4}$$

Subtracting gives $\cosh^2 \phi - \sinh^2 \phi = \frac{2 - (-2)}{4} = 1$, and adding gives $\cosh^2 \phi + \sinh^2 \phi = \frac{2e^{2\phi} + 2e^{-2\phi}}{4} = \cosh^2(2\phi)$.

- 1.2. Define the *rapidity* ϕ in terms of the velocity $\beta = v/c$ as follows:

$$\tanh \phi = \beta. \quad (3)$$

(In other words, $\phi = \tanh^{-1} \beta$ where $\tanh^{-1}$ is the inverse function to $\tanh$, just like $\tan^{-1}$ is the inverse function to the ordinary trig function $\tan$.) Show that in terms of rapidity, we can express the Lorentz transformation parameters γ and $\gamma\beta$ as

$$\gamma = \cosh \phi, \quad \gamma\beta = \sinh \phi. \quad (4)$$

The easiest way to do this is just to write $\tanh \phi$ in terms of exponentials.

We start from $\beta = \tanh \phi = \frac{e^\phi - e^{-\phi}}{e^\phi + e^{-\phi}}$. Now, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$, so let's consider

$$1 - \beta^2 = 1 - \left(\frac{e^\phi - e^{-\phi}}{e^\phi + e^{-\phi}} \right)^2 = 1 - \frac{e^{2\phi} + e^{-2\phi} - 2}{e^{2\phi} + e^{-2\phi} + 2} = \frac{4}{e^{2\phi} + e^{-2\phi} + 2} = \frac{1}{\cosh^2 \phi}.$$

Thus

$$\gamma = \frac{1}{\sqrt{\frac{1}{\cosh^2 \phi}}} = \cosh \phi,$$

where we can take the positive root because we know $\gamma \geq 1$. Now it is easy to compute $\gamma\beta = \cosh \phi \tanh \phi = \cosh \phi (\sinh \phi / \cosh \phi) = \sinh \phi$.

1.3. The Lorentz transformation matrix for a boost along the x-axis can therefore be written

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} \cosh \phi & \sinh \phi \\ \sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}. \quad (5)$$

Note how similar this looks to a rotation matrix! The essential difference is the absence of a minus sign and the replacement of ordinary trig functions with their hyperbolic cousins. Using the first identity you derived in question 1.1, show that this Lorentz boost matrix preserves the invariant interval:

$$(ct)^2 - x^2 = (ct')^2 - x'^2. \quad (6)$$

$$\begin{aligned} (ct)^2 - x^2 &= (ct' \cosh \phi + x' \sinh \phi)^2 - (ct' \sinh \phi + x' \cosh \phi)^2 \\ &= (ct')^2 (\cosh^2 \phi - \sinh^2 \phi) + x'^2 (\sinh^2 \phi - \cosh^2 \phi) + 2x' ct' (\cosh \phi \sinh \phi - \sinh \phi \cosh \phi) \\ &= (ct')^2 - x'^2 \end{aligned}$$

1.4. Another extremely convenient fact about rapidities is that they make the velocity addition formula look easy. Show that if $\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}$, then $\beta = \tanh(\phi_1 + \phi_2)$ with $\beta_1 = \tanh \phi_1$ and $\beta_2 = \tanh \phi_2$. In other words, velocity addition can be equivalently stated as *rapidities add*: $\phi = \phi_1 + \phi_2$. This is exactly the same thing that happens with 2-dimensional rotations!

$$\begin{aligned} \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} &= \frac{\tanh \phi_1 + \tanh \phi_2}{1 + \tanh \phi_1 \tanh \phi_2} \\ &= \frac{\frac{e^{\phi_1} - e^{-\phi_1}}{e^{\phi_1} + e^{-\phi_1}} + \frac{e^{\phi_2} - e^{-\phi_2}}{e^{\phi_2} + e^{-\phi_2}}}{1 + \frac{e^{\phi_1} - e^{-\phi_1}}{e^{\phi_1} + e^{-\phi_1}} \frac{e^{\phi_2} - e^{-\phi_2}}{e^{\phi_2} + e^{-\phi_2}}} \\ &= \frac{e^{\phi_1 + \phi_2} - e^{-\phi_1 - \phi_2} + e^{\phi_1 - \phi_2} - e^{-\phi_1 + \phi_2} + e^{\phi_1 + \phi_2} - e^{-\phi_1 - \phi_2} - e^{\phi_1 - \phi_2} + e^{-\phi_1 + \phi_2}}{e^{\phi_1 + \phi_2} + e^{-\phi_1 - \phi_2} + e^{\phi_1 - \phi_2} + e^{-\phi_1 + \phi_2} + e^{\phi_1 + \phi_2} + e^{-\phi_1 - \phi_2} - e^{\phi_1 - \phi_2} - e^{-\phi_1 + \phi_2}} \end{aligned}$$

In the last line we have canceled the common denominator. The mess of exponentials simplifies because the unwanted terms with $\phi_1 - \phi_2$ cancel, and we have

$$\frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = \frac{2e^{\phi_1 + \phi_2} - 2e^{-\phi_1 - \phi_2}}{2e^{\phi_1 + \phi_2} + 2e^{-\phi_1 - \phi_2}} = \tanh(\phi_1 + \phi_2).$$

- 1.5. Finally, rapidities are useful for understanding Minkowski diagrams. Show using the second of your identities derived in 1.1 that the angle θ between the ct and ct' axes (and also the x and x' axes) satisfies $\tan \theta = \tanh \phi$, and that the length of a unit on the S' axes is $\sqrt{\cosh 2\phi}$. Use the Taylor expansions of e^x and $\sqrt{1+x}$ to show that, to linear order in ϕ , units in S are the same length as units in S' .

Since we already have that $\beta = \tan \theta$ from lecture, and $\beta = \tanh \phi$ as the definition of rapidity, the relation $\tan \theta = \tanh \phi$ is trivial. Since $(x', ct') = (0, 1)$ corresponds to $(x, ct) = (\gamma\beta, \gamma)$, the length of a unit on the ct' axis is $\sqrt{(\gamma\beta)^2 + \gamma^2} = \sqrt{\sinh^2 \phi + \cosh^2 \phi} = \sqrt{\cosh 2\phi}$ by the second identity in question 1.1. An identical calculation holds for the x' axis, using $(x', ct') = (1, 0) \Leftrightarrow (x, ct) = (\gamma, \gamma\beta)$. To perform the Taylor expansion, we proceed as follows:

$$\begin{aligned} \cosh 2\phi &= \frac{1}{2} \left(e^{2\phi} + e^{-2\phi} \right) = \frac{1}{2} \left(\left(1 + 2\phi + \frac{(2\phi)^2}{2!} + \mathcal{O}(\phi^3) \right) + \left(1 - 2\phi + \frac{(-2\phi)^2}{2!} + \mathcal{O}(\phi^3) \right) \right) \\ &= 1 + 2\phi^2 + \mathcal{O}(\phi^3) \end{aligned}$$

We then use the Taylor series for $\sqrt{1+x}$ to obtain

$$\sqrt{\cosh 2\phi} = \sqrt{1 + 2\phi^2 + \mathcal{O}(\phi^3)} = 1 + \frac{1}{2}(2\phi^2) + \mathcal{O}(\phi^3) = 1 + \phi^2 + \mathcal{O}(\phi^3).$$

So to linear order in ϕ , $\sqrt{\cosh 2\phi} \approx 1$, and the leading corrections are at order ϕ^2 .

Part 2: Commutation relations for Lorentz boosts

Now that we understand boosts along the x -axis, let's bring back the boosts along the other coordinate axes as well. Keeping with the analogy to rotations, let's write these boosts in terms of rapidities:

$$\Lambda_x(\phi) = \begin{pmatrix} \cosh \phi & \sinh \phi & 0 & 0 \\ \sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_y(\phi) = \begin{pmatrix} \cosh \phi & 0 & \sinh \phi & 0 \\ 0 & 1 & 0 & 0 \\ \sinh \phi & 0 & \cosh \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_z(\phi) = \begin{pmatrix} \cosh \phi & 0 & 0 & \sinh \phi \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \phi & 0 & 0 & \cosh \phi \end{pmatrix}. \quad (7)$$

(You should compare Λ_x and Λ_y to the analogous matrices in Discussion 5 and make sure you understand where the nonzero entries go.)

- 2.1. In exact analogy to Part 2 of last week's Discussion 6, Taylor-expand the entries of $\Lambda_x(\epsilon)$, $\Lambda_y(\epsilon)$, and $\Lambda_z(\epsilon)$ to find $\Lambda_x(\epsilon) \approx I + \epsilon K_x$, $\Lambda_y(\epsilon) \approx I + \epsilon K_y$, and $\Lambda_z(\epsilon) \approx I + \epsilon K_z$, where I is now the 4×4 identity matrix and

$$K_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (8)$$

These are pretty easy to remember: the only nonzero entries are in the zeroth row and column, in the position labeled by the subscript (K_x has the 1's in the x component, K_y has the 1's in

the y component, and K_z has the 1's in the z component). To do the Taylor expansion, you can simply expand the exponentials in the definition of $\sinh$ and $\cosh$ as you did in question 1.5 of Part 1 above.

The Taylor expansions of the hyperbolic trig functions are, to linear order,

$$\begin{aligned}\cosh \phi &= \frac{e^\phi + e^{-\phi}}{2} = \frac{(1 + \phi + \mathcal{O}(\phi^2)) + (1 - \phi + \mathcal{O}(\phi^2))}{2} = 1 + \mathcal{O}(\phi^2) \\ \sinh \phi &= \frac{e^\phi - e^{-\phi}}{2} = \frac{(1 + \phi + \mathcal{O}(\phi^2)) - (1 - \phi + \mathcal{O}(\phi^2))}{2} = \phi + \mathcal{O}(\phi^2)\end{aligned}$$

Therefore, in the infinitesimal Lorentz transformation matrices, $\cosh \epsilon \rightarrow 1$ and $\sinh \epsilon \rightarrow \epsilon$. For example,

$$\Lambda_x(\epsilon) = \begin{pmatrix} 1 & \epsilon & 0 & 0 \\ \epsilon & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \epsilon \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = I + \epsilon K_x$$

- 2.2. Compute the commutator $[K_x, K_y]$. Is it equal to K_z ? If not, this means that *boosts along different axes do not form a group!*¹

$$\begin{aligned}[K_x, K_y] &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.\end{aligned}$$

This is definitely not the same as K_z !

- 2.3. Rotations of the spatial coordinates do not change the time coordinate, by definition. This means that the rotation matrices from last week (which are 3×3 matrices) can be embedded in 4×4 matrices, for example

$$R_z(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (9)$$

¹Strictly speaking, we should actually check that $[K_x, K_y]$ is not equal to a linear combination of any of the three generators K_x, K_y, K_z . This is indeed the case.

Let this matrix act on a vector (ct', x', y', z') and show that the transformation is $ct = ct', x = x' \cos \theta - y' \sin \theta, y = x' \sin \theta + y' \cos \theta, z = z'$, as expected for a rotation around the z -axis.

Doing the matrix-vector multiplication should be straightforward, but we can interpret the result: $ct = ct'$ and $z = z'$ come from the fact that the first and last rows have 1 on the diagonal and 0's elsewhere. The 2×2 block is identical to $R(\theta)$ as studied in Discussion 6, and its placement in the matrix means that it acts on the x and y coordinates.

- 2.4. Compute the 4×4 analogue of the infinitesimal rotation J_z from Discussion 6 by Taylor-expanding the 4×4 matrix R_z above, and show that $[K_x, K_y]$ as computed in question 2.2 is equal to $-J_z$. The interpretation is that *two boosts along two different axes involve a rotation of the coordinate systems*. You can't have Lorentz transformations without rotations!

Using $\cos \epsilon \approx 1$ and $\sin \epsilon \approx \epsilon$, and revisiting the calculation in Discussion 6,

$$R_z(\epsilon) \approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \epsilon \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so we can identify $J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. This is indeed the negative of the matrix computed

in 2.2, so $[K_x, K_y] = -J_z$.

As with the rotation group, you can check the rest of the commutators to show that the commutator of a K and a J gives something proportional to K . This means that any combination of infinitesimal rotations (about any axis) and boosts (along any axis) is also a combination of a rotation and a boost. The combination of boosts and rotations is called the *Lorentz group*. The rotation group $SO(3)$ is a subgroup of the full Lorentz group.

Part 3: The Lorentz group $SO(3,1)$

Recall in lecture that we introduced the metric

$$g = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad (10)$$

which lets us encode the invariant interval as $\Delta s^2 = g_{\mu\nu} \Delta x^\mu \Delta x^\nu$. Just as orthogonal matrices preserve the length of a vector, Lorentz matrices preserve the invariant interval, which can be thought of as the relativistic “length” of the displacement 4-vector Δx^μ . The mathematical expression of this fact is the equation

$$g = \Lambda^T g \Lambda \quad (11)$$

which can be taken as a *definition* of the Lorentz group. If we replaced g with the 4×4 identity matrix, we would simply get $I = \Lambda^T \Lambda$, which is satisfied by an orthogonal matrix. But since g has 3 minus signs and 1 plus sign, we call the Lorentz group $SO(3,1)$. (Sometimes you'll see this written as $SO(1,3)$; this is convention-dependent exactly like the sign of the metric.)

3.1. Show that the Lorentz matrix Λ_x satisfies Eq. (11).

Since $\Lambda_x(\phi)^T = \Lambda_x(\phi)$, we have

$$\begin{aligned}
\Lambda_x^T g \Lambda &= \begin{pmatrix} \cosh \phi & \sinh \phi & 0 & 0 \\ \sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \cosh \phi & \sinh \phi & 0 & 0 \\ \sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \cosh \phi & \sinh \phi & 0 & 0 \\ \sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh \phi & \sinh \phi & 0 & 0 \\ -\sinh \phi & -\cosh \phi & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} \cosh^2 \phi - \sinh^2 \phi & \cosh \phi \sinh \phi - \sinh \phi \cosh \phi & 0 & 0 \\ \sinh \phi \cosh \phi - \cosh \phi \sinh \phi & \sinh^2 \phi - \cosh^2 \phi & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
&= g
\end{aligned}$$

3.2. Compute $\Lambda_x(\phi_1)\Lambda_y(\phi_2)$ in the rapidity parameterization and show that it *also* satisfies Eq. (11).

Rather than multiply the matrices directly, it's actually easier to use the result from 3.1:

$$(\Lambda_x(\phi_1)\Lambda_y(\phi_2))^T g (\Lambda_x(\phi_1)\Lambda_y(\phi_2)) = \Lambda_y(\phi_2)^T (\Lambda_x(\phi_1)^T g \Lambda_x(\phi_1)) \Lambda_y(\phi_2) = \Lambda_y(\phi_2)^T g \Lambda_y(\phi_2).$$

The computation of $\Lambda_y(\phi_2)^T g \Lambda_y(\phi_2)$ is essentially identical to 3.1, and yields g (as it must, because Λ_y is a boost which we know preserves the invariant interval.) Therefore $\Lambda_x(\phi_1)\Lambda_y(\phi_2)$ satisfies the defining relationship for the Lorentz group.

At last, we have an answer to our question from Discussion 5: $\Lambda_x\Lambda_y$ is indeed a member of the Lorentz group because it preserves the invariant interval, but it is *not* a boost: it is a combination of a boost and a rotation.